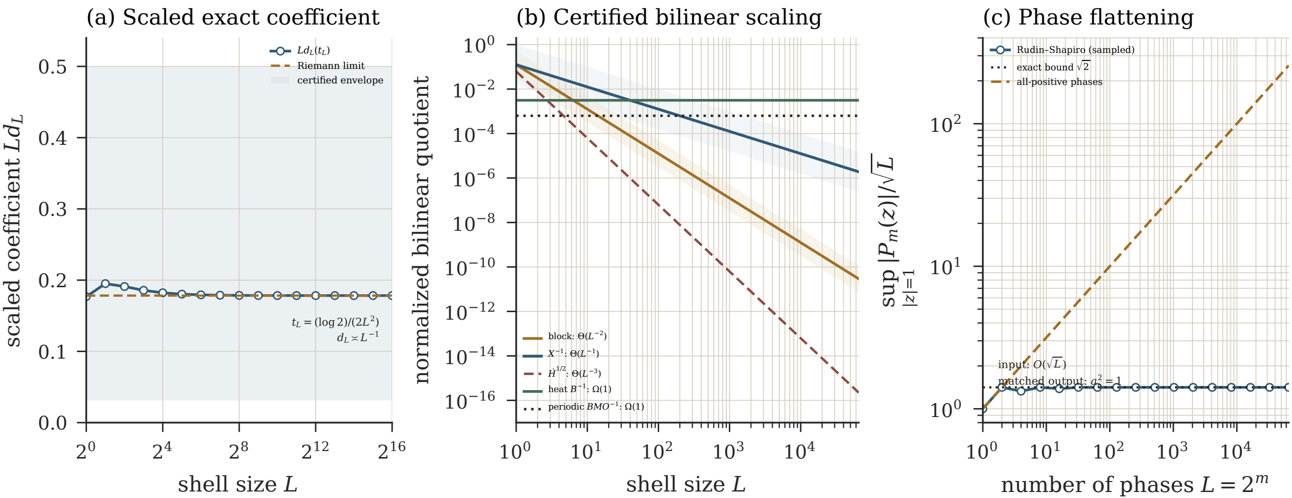
Exact Duhamel denominator: shell gain versus critical saturation



Exact output: $d_L(t) = e^{-t}\sum_{N=L}^{2L-1}(1-e^{-2N^2 t})/(2N^2)$. At $t_L = (\log 2)/(2L^2)$, $1/(32L) \le d_L(t_L) \le 1/(2L)$ for every L.

Formal scope: 8,390,656 packet modes, Rudin–Shapiro length 4,194,304, 24/24 checks; exact integer and Q(sqrt(2)) audit. Curves and sampled maxima are presentation data.

Conclusion: the time integral gives a genuine heat denominator, but critical negative regularity and phase flattening can consume its shell gain. This is not norm inflation or a Navier–Stokes regularity proof.